

# ISAAC ROSENBERG

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## EDUCATION

### • Erasmus Mundus Joint Master in Astrophysics

*Master of Science in Astrophysics*

September 2025 – July 2027

Italy, Serbia, France

- Two-year joint program across University of Rome Tor Vergata, University of Belgrade and Université Côte d'Azur focused on astrophysics and space science.
- **Current grade:** 29.7 / 30

### • University of Toronto

*Honours Bachelor's of Science*

September 2021 – May 2025

Toronto, Canada

- Astronomy and Astrophysics Specialist, Mathematics Minor.
- **Grade:** 3.85 / 4.00

## PUBLICATIONS

1. **Isaac Rosenberg**, Martine Lokken, Renée Hložek, Weiguang Cui, Sara Santoni, Romeel Davé (2026). [Here, There and Everywhere: How AGN Jets Affect Galaxy Cluster Environments](#). Submitted for publication in *Astronomy & Astrophysics*.

## RESEARCH EXPERIENCE

### The Impacts of AGN Feedback on Galaxy Cluster Environments

2024 - 2026

*Department of Astronomy, University of Toronto*

- Research project with Professor Renée Hložek and Dr. Martine Lokken.
- Ran hydrodynamic simulations with the 300 collaboration of galaxy clusters to understand how AGN feedback affects galaxy and cluster properties.
- Paper submitted to *Astronomy & Astrophysics*.

### Standard Siren Cosmology with Gravitational Waves

2024 - 2026

*Canadian Institute for Theoretical Astrophysics*

- Research project with Professor Maya Fishbach.
- Built a model that can simultaneously fit for the cosmological and astrophysical evolution of the binary black hole mass distribution.
- Received Summer Undergraduate Research Fellowship to continue this project over summer 2025.
- Paper in preparation.

### Summer Research, ATLAS Experiment

Summer 2024

*CERN, Switzerland*

- Worked under Professors Nikolina Ilic and Richard Teuscher on the ATLAS experiment at CERN for 12 weeks, in person.
- Investigated yearly spatial variations in jet distributions of the  $H \rightarrow ZZ$  particle decay.

## CONFERENCES, SCHOOLS & TALKS

### 1. The Three Hundred Annual Collaboration Meeting

August 2026

*La Cristalera, Spain*

Talk

Research talk on AGN feedback simulations; five days of discussions and workshops on galaxy cluster simulations.

### 2. International Conference on Physics Students

August 2026

*Rabat, Morocco*

Talk

Research talk on gravitational wave cosmology; five days of lectures and workshops across broad topics in physics.

### 3. Flatiron Institute CMB Summer School

July 2026

*New York, US*

School

Five-day workshop on CMB data analysis with the Simons Observatory.

### 4. Master in Astrophysics and Space Science Winter School

Feb 2026

*Rome, Italy*

School

Four-day workshop on machine learning and data science applications to astrophysics.

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| 5. <b>FLATs Talk Series</b><br><i>University College London, UK</i><br>Research talk on galaxy clusters.                               | January 2026<br>Talk    |
| 6. <b>Cosmology Meeting</b><br><i>Imperial College London, UK</i><br>Research talk on galaxy clusters.                                 | January 2026<br>Seminar |
| 7. <b>Simulations Meeting</b><br><i>University of Oxford, UK</i><br>Research talk on galaxy clusters.                                  | January 2026<br>Seminar |
| 8. <b>CMB – Large Scale Structure Meeting</b><br><i>University of Cambridge, UK</i><br>Research talk on galaxy clusters.               | January 2026<br>Seminar |
| 9. <b>EXPLORE Summer School</b><br><i>Toronto, Canada</i><br>Research talk on gravitational wave cosmology; winner of best talk award. | August 2025<br>Talk     |
| 10. <b>CASCA Conference</b><br><i>Halifax, Canada</i><br>Research poster and talk on galaxy clusters.                                  | June 2025<br>Poster     |
| 11. <b>STATSTRO Conference</b><br><i>Toronto, Canada</i><br>Research talk on gravitational wave cosmology.                             | May 2025<br>Talk        |

## HONORS AND AWARDS

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| • <b>Winner of Best Talk Award at EXPLORE Summer School</b><br><i>York University</i>                                                                                        | 2025                   |
| • <b>Erasmus Mundus Joint Masters Scholarship</b><br><i>Erasmus MASS program</i>                                                                                             | 2025 – 2027<br>€34,000 |
| • <b>Summer Undergraduate Research Fellowship</b><br><i>Canadian Institute for Theoretical Astrophysics</i><br>◦ Summer fellowship to research gravitational wave cosmology. | 2025<br>\$9,500        |
| • <b>CIE International Experience Award</b><br><i>Centre for International Experience, University of Toronto</i>                                                             | 2024<br>\$1,500        |
| • <b>Helen Mae Woodliffe Scholarship</b><br><i>Victoria College, University of Toronto</i>                                                                                   | 2024<br>\$1,000        |
| • <b>John Hamilton Watson Award for International Study</b><br><i>Victoria College, University of Toronto</i>                                                                | 2024<br>\$3,000        |
| • <b>Prof. William Kingston and Dr. John Kingston Scholarship</b><br><i>Victoria College, University of Toronto</i>                                                          | 2023<br>\$1,000        |
| • <b>Regents In-course Scholarship I</b><br><i>Victoria College, University of Toronto</i>                                                                                   | 2022<br>\$1,000        |
| • <b>Regents Admission Scholarship</b><br><i>Victoria College, University of Toronto</i>                                                                                     | 2021<br>\$5,000        |

## TEACHING EXPERIENCE

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| <b>Teaching Assistant</b><br><i>University of Toronto, Department of Mathematics &amp; Department of Astronomy</i>                                                                                  | 2023 - 2025 |
| • Teaching assistant for <a href="#">MAT186 (Calculus I)</a> , <a href="#">MAT187 (Calculus II)</a> , <a href="#">MAT188 (Linear Algebra)</a> , and <a href="#">AST251 (Life on Other Worlds)</a> . |             |
| • Led three tutorials per week, marked assignments, invigilated exams, and held office hours for university calculus, linear algebra, and astronomy courses with over 4,000 students combined.      |             |

## RELEVANT COURSEWORK

*\* Graduate-level courses in italics*

### Astrophysics

- Stars and Planets, Galaxies and Cosmology, Astrophysics Laboratory, *Cosmology, Stellar Astrophysics, Radiative Processes, Active Galactic Nuclei, Plasma Spectroscopy, Computational Astrobiology, Interstellar Medium, General Relativity.*

### Physics

- Classical Mechanics I & II, Electromagnetism I & II, Thermal Physics, Computational Physics, Physics Laboratory, *High Energy Physics, Quantum Mechanics.*

### Mathematics and Computer Science

- Analysis, Multivariable Calculus, Linear Algebra I & II, Differential Equations, Introduction to Computer Programming, Neural Networks, *Complex Analysis, ODEs & PDEs.*

## SKILLS

- **Programming Languages:** Python, C, C++, Perl, Java, MATLAB, ROOT
- **Hydrodynamic Simulations (and related tools):** GIZMO, GADGET, CAESAR, YT, DisPerSE
- **Python Packages:** Numpy, Scipy, Matplot, Pandas, Astropy, PyTorch, scikit-learn

## PROFESSIONAL MEMBERSHIPS

- **The Three Hundred Collaboration** *October, 2024 - Present*

## ADDITIONAL INFORMATION

**Languages:** English (native), French (intermediate), Russian (intermediate), German (beginner)

**Interests:** Guitar, singing, world history, baseball, travel, chess, politics, music, football.